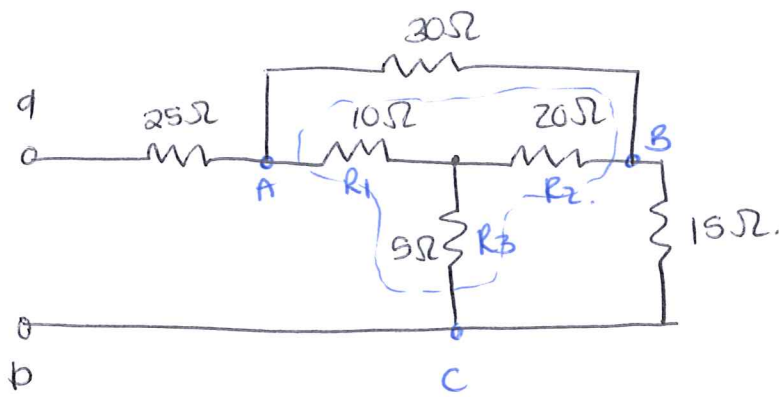
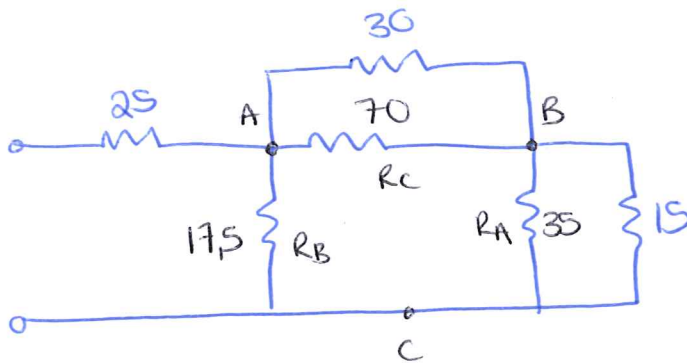


b)

(2)



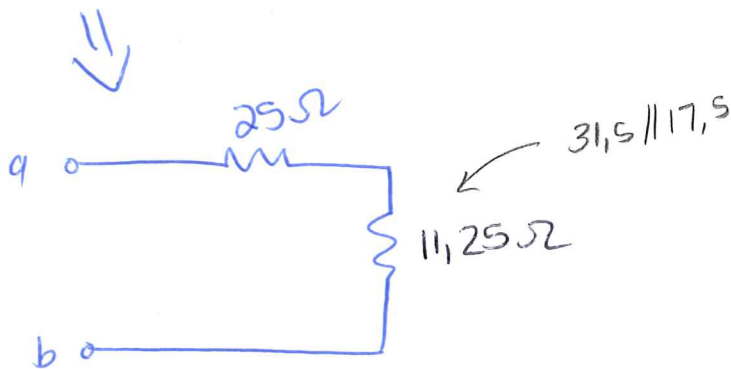
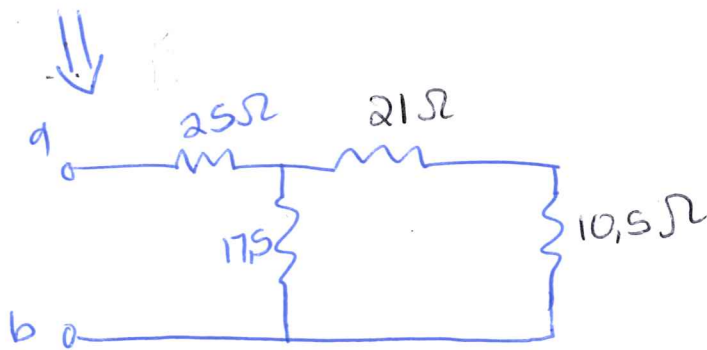
⇓ Y → Δ



$$R_A = \frac{(10 \times 20) + (20 \times 5) + (5 \times 10)}{10}$$

$$R_B = \frac{350}{20}$$

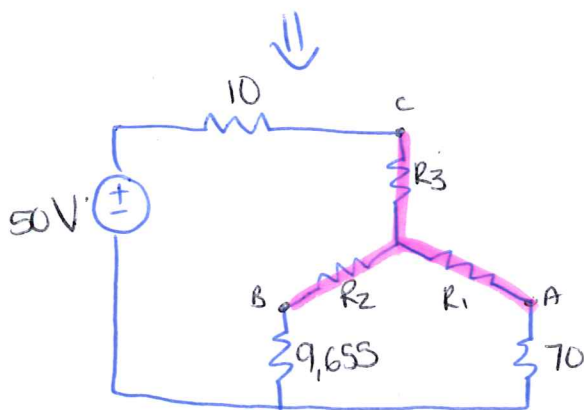
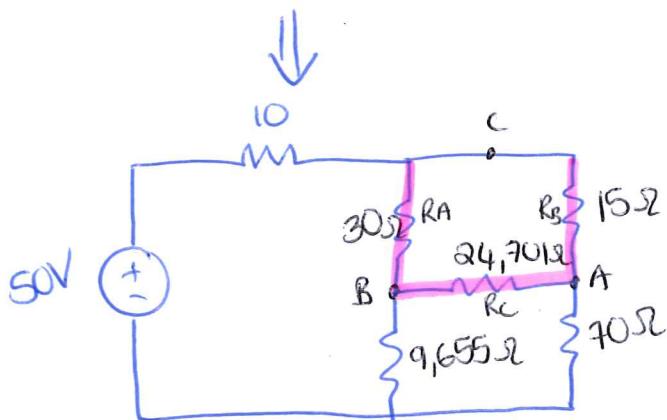
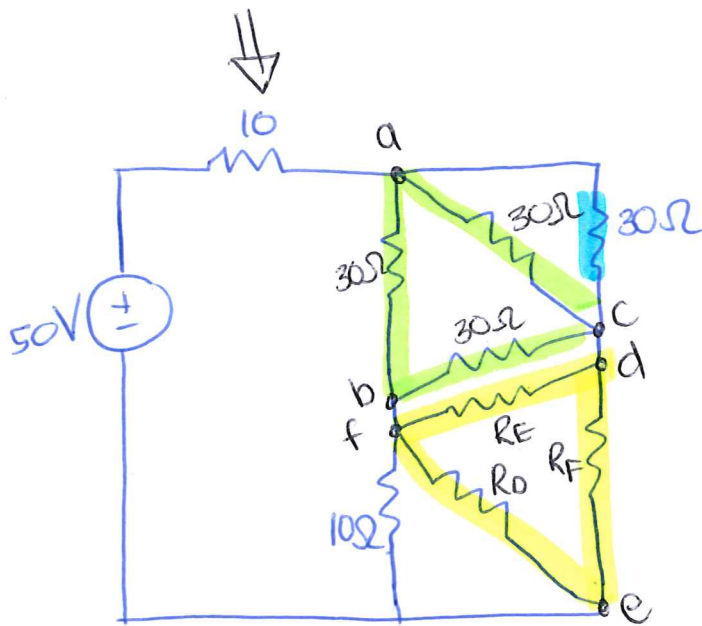
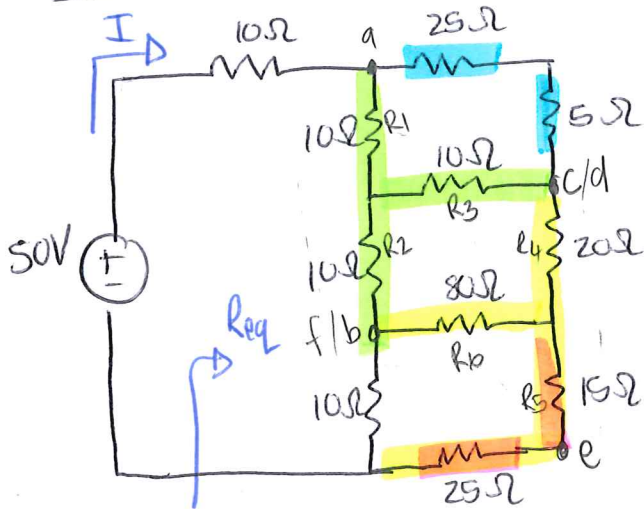
$$R_C = \frac{350}{5}$$



$$\therefore R_{ab} = 25 + 11,25 = \underline{\underline{36,25 \Omega}}$$

(4)

2.57

Determine R_{eq} and I .

$$* 600 = (20 \times 80) + (20 \times 40) + (40 \times 80)$$

$$R_D = \frac{600^*}{20} = 280 \Omega$$

$$R_E = \frac{600}{40} = 140 \Omega$$

$$R_F = \frac{600}{80} = 70 \Omega$$

$$* 30 \parallel R_E = 24.701 \Omega$$

$$* 30 \parallel 30 = 15 \Omega$$

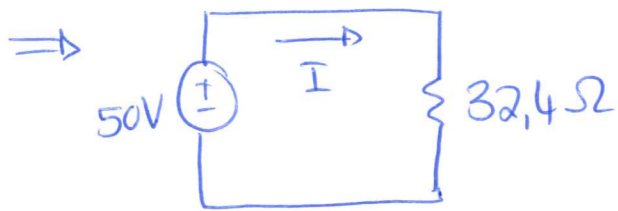
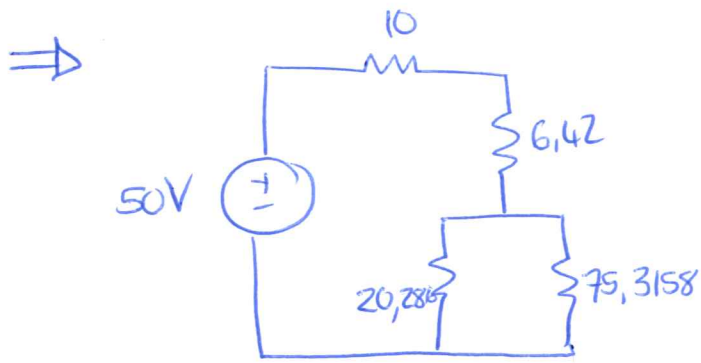
$$* 10 \parallel R_D = 9.655 \Omega$$

$$R_1 = \frac{15 \times 24.701}{69.701} = 5.3158 \Omega$$

$$R_2 = \frac{30 \times 24.701}{69.701} = 10.6316 \Omega$$

$$R_3 = \frac{30 \times 15}{69.701} = 6.4561 \Omega$$

$$69.701 = 30 + 15 + 24.701$$



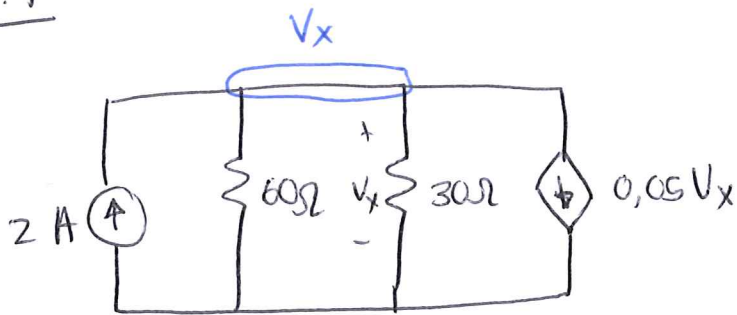
$$\therefore I = \frac{50}{32,4}$$

$$= 1,54 \text{ A}$$

$$R_{eq} = 32,4 \Omega.$$

3.7

8

Find V_x

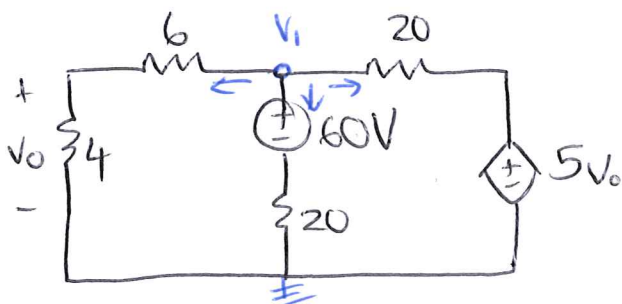
@ node x: $\frac{V_x}{60} + \frac{V_x}{30} + 0.05V_x = 2$

$$V_x + 2V_x + 3V_x = 120$$

$$6V_x = 120$$

$$\underline{V_x = 20 \text{ V}}$$

3.8

Find V_o

$$\frac{V_1 - 0}{6+4} + \frac{(V_1 - 60) - 0}{20} + \frac{V_1 - 5V_o}{20} = 0$$

$$4V_1 - 5V_o = 60 \quad \text{--- (1)}$$

$$\begin{aligned} * V_o &= \left(\frac{4}{4+6} \right) V_1 \quad \text{[voltage division]} \\ &= \frac{2}{5} V_1 \quad \text{--- (2)} \end{aligned}$$

∴ (2) in (1)

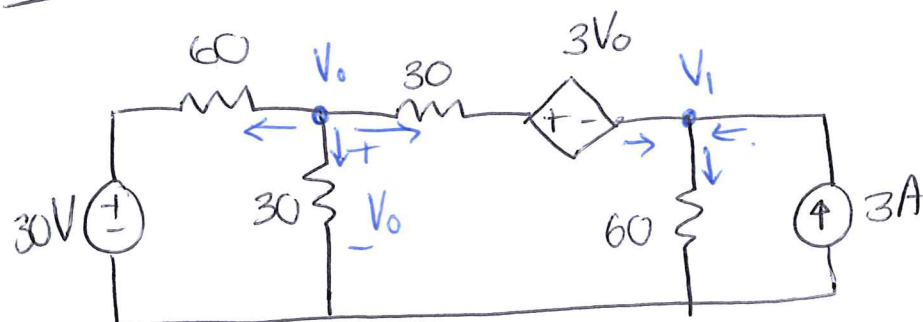
$$\Rightarrow 4V_1 - 5\left(\frac{2}{5}V_1\right) = 60$$

$$V_1 = 30 \text{ V}$$

$$\therefore \underline{V_o = \frac{2}{5}(30) = 12 \text{ V}}$$

3.23

(11)



Use nodal analysis to find V_0 .

@ node 0: $\frac{V_0 - 30}{60} + \frac{V_0 - 0}{30} + \frac{V_0 - (3V_0 + V_1)}{30} = 0$

$$V_0 - 30 + 2V_0 + 2V_0 - 6V_0 - 2V_1 = 0$$

$$-V_0 - 2V_1 = 30$$

@ node 1:

$$\Rightarrow V_0 = -30 - 2V_1 \quad \text{--- (1)}$$

$$\frac{V_0 - (3V_0 + V_1)}{30} + 3 = \frac{V_1}{60}$$

$$2V_0 - 6V_0 - 2V_1 + 180 = V_1$$

$$-4V_0 - 3V_1 = -180 \quad \text{--- (2)}$$

① in ②

$$\therefore -4(-30 - 2V_1) - 3V_1 = -180$$

$$120 + 8V_1 - 3V_1 = -180$$

$$5V_1 = -300$$

$$V_1 = -60 \text{ V}$$

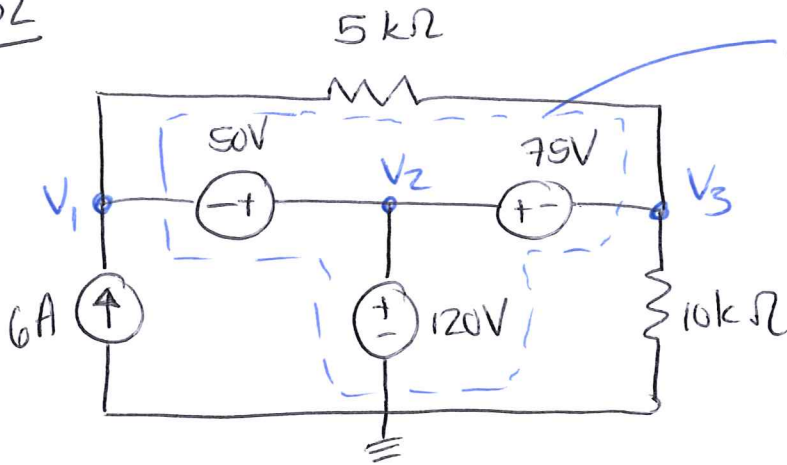
$$\therefore V_0 = -30 - 2(60)$$

$$= 90 \text{ V}$$

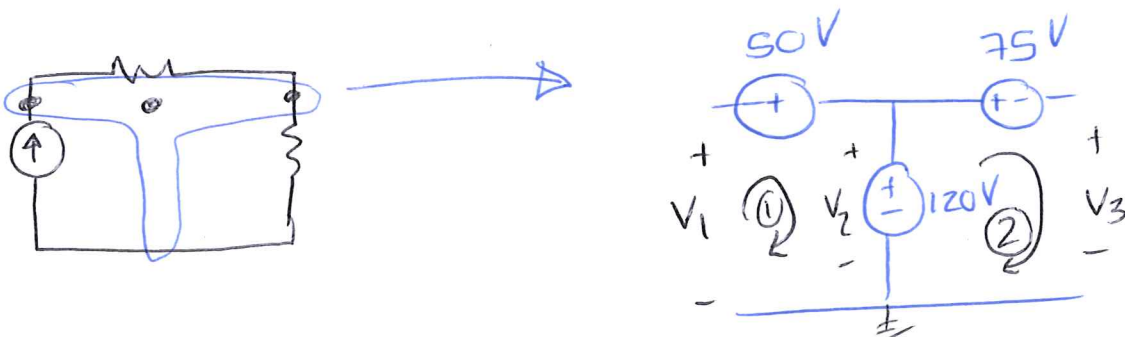
→

3.32

(12)

Find V_1, V_2, V_3 .

↳ "Supernode"

* $V_2 = 120V$ (evident from original fig)* KVL @ ①:

$$-V_1 - 50 + 120 = 0$$

$$\underline{V_1 = 70V}$$

* KVL @ ②:

$$-120 + 75 + V_3 = 0$$

$$\underline{V_3 = 45V}$$

$V_1 = 70V$
$V_2 = 120V$
$V_3 = 45V$

Tutorial 2

(Answers)

2.51 a) $R_{eq} = 9,23 \Omega$ *

b) $R_{eq} = 36,25 \Omega$

2.56 $V = 42,18 V$ *

2.57 $R_{eq} = 32,4 \Omega$

$I = 1,54 A$

3.2 $V_1 = 0 V$ *

$V_2 = 12 V$

3.4 $i_1 = 3 A$ * ; $i_2 = 6 A$; $i_3 = i_4 = -0,5 A$

3.7 $V_x = 20 V$

3.8 $v_0 = 12 V$

3.15 $i_0 = 29,45 A$ *

$P_1 = 144,6 W$

$P_2 = 129,6 W$

$P_3 = 12 W$

3.17 $i_0 = 1,73 A$ *

3.23 $V_0 = 90 V$

3.32 $V_1 = 70 V$ $V_2 = 120 V$ $V_3 = 45 V$